Serial Number _____

CYCLE LIFE TESTER

MODEL LCV (Multiple Power Module Cabinet)

HARDWARE OPERATION MANUAL

The purpose of this manual is to assist the operator in the proper installation, operation, and maintenance of the Bitrode Model LCV Cycle Life Tester. This information has been carefully checked and is believed to be entirely reliable. However, no responsibility is assumed for inaccuracies. Should any discrepancies or questions occur, the customer may contact Bitrode Corporation for assistance.

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TABLE OF CONTENTS

REVISIONS II
TABLE OF CONTENTS III
1.0 INTRODUCTION 1
2.0 OPERATOR SAFETY 3
2.1 Warning Statements 3
2.2 Safety Considerations 3
3.0 INSTALLATION 4
3.1 Inspection Upon Receipt 4
3.2 Determining Proper Location 4
3.2.1 Lifting/Moving 4
3.3 Main Cabinet Connections 4
3.3.1 Power Line 4
3.3.2 Bitrode Communications Adapter (BCA) Cable 5
3.3.3 Battery Cables 6
3.3.4 Inputs/Outputs (I/O) 6
3.3.4.1 Thermocouple Inputs 7
3.3.4.2 STM Temperature Inputs 7
3.3.4.3 Reference Voltage Inputs 7
3.3.4.4 Pressure Transducer Inputs 7
3.3.4.5 Digital or Relay I/O 8
3.3.4.6 Thermistor Inputs 8
3.4 LCV Circuit Address Selection 8
3.5 Bitrode Communications Adapter (BCA) Setup 8
4.0 HARDWARE OPERATION 9
4.1 Initial Power Up 9
4.1.1 LCV Cabinet 9
4.1.2 Bitrode Communications Adapter (BCA) 9
4.2 LCV Cabinet 9
4.2.1 Cabinet Description 9
4.2.2 Parallel Circuit Capability 10
4.2.3 Assignable Data Channel Capability (Optional) 10
4.2.4 Isolated High Current Heatsink Description 10
4.2.5 Controls, Indicators and Jacks 11
4.3 Battery Connection 11
5.0 CALIBRATION & PREVENTIVE MAINTENANCE 13
5.1 Calibration 13
5.1.1 Equipment Required 13
5.1.2 Calibration Procedure 13
5.2 Preventive Maintenance 15
6.0 SPECIFICATIONS 16

1.0 INTRODUCTION

The Bitrode Model LCV is a multiple circuit, microprocessor controlled, cycle life tester. This tester is capable of performing standard or fully customized charge and discharge cycles on storage batteries. Each circuit can be programmed to cycle batteries precisely according to SAE, DIN, BCI, or MIL spec. requirements, or to perform reserve capacity and charge acceptance tests. The unit will also perform custom tests to satisfy the user's own methods of analyzing storage battery potential.

The VisuaLCN Software package, designed to be used with any IBM PC or compatible computer, allows for complete management of all LCV circuit operations. The menu driven package contains all programming, data collection, and monitoring functions necessary to take full advantage of the system's capabilities.

A typical LCV system can be equipped with multiple cabinets, with one or more circuits per cabinet, for a total of up to 99 circuits connected to a single common communications network. All communication and programming of the circuits takes place on a single communications cable. Network communication is managed by the Bitrode Communications Adapter (BCA). The RS-232 serial port on the BCA provides for connection to the host computer system. Larger systems may have more than one network connected to the host computer allowing communication with more than 99 circuits.

Each circuit of the LCV system can be programmed and operated independently so that several different types of cycles or tests (one per circuit) can be run simultaneously. The circuits can charge and discharge at any current and voltage within the limits listed in the Specifications section of this manual. A summary of the programming capabilities is listed in the LCV Cabinet Description section.

The Model LCV provides a parallel circuit capability, which allows multiple circuits in a single cabinet to be combined/re-combined as needed to provide different maximum current configurations. Also, an assignable data channel option allows for input channels, installed in the cabinet, to be assigned/re-assigned to different circuits as needed. The combination of these two options provides an incredibly flexible battery cycling capability.

If a circuit is turned off or a power failure occurs, loss of programs and data is averted by the battery-backed memory within each cabinet. If a power failure occurs while a test is running, the test will resume as soon as power returns. This feature allows the unit to operate unobserved for long periods of time. As an option, the system can be designed to not resume after a power failure. This option is beneficial if critical tests can not be resumed after power fail.

The LCV unit contains protection from high-level external interference or other disruptive events that could result in an improperly operating microprocessor. This protection includes several run-time monitors (in hardware and firmware) that constantly check to see if the microprocessor is operating as designed. The LCV is designed to reset, then halt, if any of these checks fail. This is different from operation after a power failure, where the LCV resumes testing after power returns. The LCV is designed to halt after a disruptive event, to protect the operator and equipment from potentially dangerous conditions, and to protect the integrity of the test data taken to that point.

The Bitrode Model LCV has been designed to address test flexibility requirements present in the battery manufacturing industry today and is thus an incredibly versatile piece of cycling equipment. Its design also provides for service and maintenance, which is important in a device that may operate on a continuous basis. This unit should provide years of beneficial and trouble free service.

2.0 OPERATOR SAFETY

2.1 Warning Statements

NOTE statements identify practices that must be followed to provide full and repeatable functionality.

CAUTION statements identify conditions or practices that could result in damage to this equipment or other property.

DANGER statements identify conditions or practices that could result in personal injury or loss of life.

2.2 Safety Considerations

DANGER/CAUTION: The following items are safety considerations:

Power Source: Make certain that the unit is connected to the correct AC power source. Ensure that the input line voltage matches the voltage specified on the nameplate.

Proper Grounding: The unit should be provided with a proper ground connection to the cabinet. Failure to provide a ground path could cause all accessible conductive parts to render an electric shock.

Fuse Replacement: To avoid a fire hazard replace fuses only with ones of the same type, voltage rating, and current rating.

Operating in Explosive Atmospheres: To avoid the possibility of an explosion, do not operate this equipment in an explosive atmosphere without specific certification for such operation.

Operation with Covers Removed: To avoid the possibility of electric shock, DO NOT operate this unit with its covers removed.

Heatsink Cooling Fans: The heatsink cooling fans operate at high speeds and are extremely dangerous. Keep hands and objects clear of the fan blades when power is applied to the cabinet.

Heated Exhaust Air: Some units produce high temperature exhaust air during discharge. Warning labels are located near the heated air exhaust vents in these cases.

Battery Cables: Disconnecting the battery cables under load is a potentially hazardous act, easily leading to an explosion-causing arc on a battery post. It is advisable to avoid placing a disconnecting means in the battery cable lines near the batteries themselves.

Service and Maintenance: Never remove or insert the circuit board assemblies or other components with the power on. Service work should only be performed with the power off and batteries disconnected. Also, SERVICE WORK SHOULD ONLY BE PERFORMED BY QUALIFIED PERSONNEL.

3.0 INSTALLATION

3.1 Inspection Upon Receipt

Upon receipt of this unit conduct a thorough inspection for damage that may have occurred during shipment. Examine the exterior and interior of the unit for any evidence of physical damage. Should any damage be detected do not attempt to operate the unit and contact the transportation carrier immediately. The LCV system may be composed of more than one cabinet and may also include a Bitrode Communications Adapter (BCA) and other related parts. Thorough inspection of all items is recommended.

3.2 Determining Proper Location

The location of this equipment is a matter deserving special consideration. This unit requires a certain amount of cooling air to flow through the components for proper operation. The unit must not be confined to any area lacking adequate ventilation. Also, selecting a location with a corrosive atmosphere will shorten the life of the equipment.

DANGER/CAUTION: Some units produce high temperature exhaust air during discharge. Warning labels are located near the heated air exhaust vents in these cases.

CAUTION: When installing this unit the ventilation openings must not be restricted. Premature failure of critical components will occur if the openings are blocked. Leave at least 2 feet between the cabinet ventilation openings and any wall.

CAUTION: This unit contains microprocessor hardware which is highly susceptible to corrosion damage. Shorter operating life can be expected from this equipment if it is exposed to a corrosive atmosphere.

3.2.1 Lifting/Moving

Most LCV cabinets are equipped with casters to allow easy movement from place to place. When it becomes necessary to lift an LCV cabinet, suitable lifting/moving means (such as slings or a forklift) should be placed around/under the cabinet so that the lifting force is EVENLY applied to the bottom of the cabinet, keeping the cabinet balanced.

3.3 Main Cabinet Connections

3.3.1 Power Line

The power line connection to the unit should be made in accordance with local building codes. If such codes are not applicable in your area, proper electrical installation of the equipment can be achieved by following the procedures of the National Electrical Code. A handbook of such is available from McGraw-Hill Company in New York or from local bookstores.

It is recommended that an electrical disconnect switch be installed to feed the unit. Select the fuse size for this switch in accordance with the input amperes on the serial number tag. The serial number tag also specifies the correct line voltage and phase required by the unit.

DANGER: The input line voltage is typically 240 to 480 volts. Exercise extreme caution when installing the input lines or servicing this equipment. Turn off the power to the unit before making contact with any of the high voltage connections.

CAUTION: Failure to supply the correct voltage and phase could result in serious damage to the unit. Also ensure that the input line is sized in accordance with the input amperes stated on the serial number tag.

The power line connects to the LCV cabinet at the input terminal block or the EMI filter (on CE units) located inside the top of the cabinet. The input AC line should enter the cabinet through the hole provided in the top just above the terminal block or EMI filter. When installing the power line, make certain a proper ground lead is also installed in the event a ground connection through conduit is not available. The ground lead should be connected to the ground lug provided adjacent to the terminal block or to the EMI filter ground stud.

DANGER: Failure to provide a ground connection could result in an electrical shock when any conductive surface of the unit is touched.

NOTE: The measurement techniques used in the LCV design are so precise that the results may be affected by the presence of high-level external interference in the area where the LCV system is located. Deviations of up to 1% of full-scale readings of voltage and current may be seen under high levels of interference. To assure reliable operation in the presence of interference, the unit must be installed and operated in accordance with the instructions in this manual. The unit should be located as far away from potential interference sources as possible; in particular, sources of radio-frequency (RF) energy such as cellular phones or other portable communications equipment must not be located (during use) within three meters (ten feet) of the LCV cabinet.

3.3.2 Bitrode Communications Adapter (BCA) Cable

An entry hole, located on the top of the LCV cabinet, allows for the communications network to be connected to the cabinet interface adapter PCB. This cabinet interface adapter PCB acts as the interface to the cabinet's MPU PCB.

A length of shielded 2-pair cable is provided with the unit. Inside the LCV cabinet, connect one end of this cable to either of the outside connectors on the cabinet interface adapter PCB. Connect the other end of this cable to the BCA or to another LCV cabinet that is already connected to the network. The other outside connector on the cabinet interface adapter PCB is used to attach the network to yet another cabinet or to terminate the line. If this LCV cabinet is the last one or the only one connected to the network, the unused connector should be configured as a terminator (See BCA Network Interconnect drawing, GS10016-01).

The communications cable can be routed in any convenient manner. However, the cable should be routed away from high voltage power lines and other sources of electromagnetic interference. The total length of the communications cable should not exceed 1000 feet. If it becomes necessary to change the length of cables, the existing cable can be easily removed from/reinserted into the screw terminals on the connectors. Additional lengths of communications cable are available from Bitrode Corporation.

NOTE: The communications cable supplied by Bitrode is a low capacitance cable. Use of higher capacitance cables over long distances can cause communication errors.

NOTE: A bad connection at any one connector can cause loss of communication with a large number of circuits.

3.3.3 Battery Cables

Each battery cable used with the LCV unit consists of a current carrying lead pair and a voltage sense lead pair. The cables may be connected directly to a nearby rack of batteries or routed through conduit to batteries in a separately enclosed area. Various length battery cables are available from Bitrode Corporation. Typically, the battery cables are supplied with the LCV unit. However, if battery leads are needed, the current carrying leads should be sized in accordance with the maximum DC current and maximum cable voltage drop listed in the specifications section of this manual.

The battery cables plug into connectors at the rear of the LCV cabinet. The cables are marked with circuit numbers. On parallel circuit capable cabinets, note that the output leads of all circuits are isolated from each other; therefore connections of multiple circuits to batteries tied in series or in parallel is allowed. Note however, that when circuits are connected to batteries in series strings, the maximum voltage differential between the connection points on the series string should not exceed the maximum circuit isolation voltage listed in the specifications section of this manual.

DANGER: All battery connections should be clean and tight. Any arcing at the connections could cause a dangerous explosion of the battery.

DANGER/CAUTION: Undersizing the battery cables may result in damage to the cable insulation. This could cause arcing at the connections thereby resulting in a dangerous battery explosion.

CAUTION: Consult the Specifications section of this manual before connecting the cables to batteries. Damage to the equipment could result if the cables are attached to batteries that exceed the voltage ratings of the unit.

NOTE: Full discharge & charge across the specified voltage range may not be achieved if the resistance of the current carrying leads is too high since the unit is designed with a maximum cable voltage drop. The maximum cable voltage drop is listed in the specifications section of this manual.

3.3.4 Inputs/Outputs (I/O)

Refer to the Specifications section for the temperature, voltage, pressure and/or digital I/O specified with this unit. For a description of how this unit's I/O is mapped to the individual circuits, see the Data Mapping Details, also provided in the Specifications section of this manual.

3.3.4.1 Thermocouple Inputs

Some units are equipped with optional thermocouple temperature inputs. In this case, thermocouple connectors are located on the I/O panel. Thermocouple probes may be attached at the panel and used to monitor the test battery temperature. The Specifications section of this manual lists the thermocouple type and temperature range. Due to the wide variety available, probes are not supplied with the unit unless specified by the customer. The probes used must be of the correct type and should be ordered to suit the particular application. The thermocouple wire should be routed away from any sources of electromagnetic interference.

3.3.4.2 STM Temperature Inputs

Some units are equipped with Bitrode's optional Semiconductor Temperature Measurement (STM) inputs as a temperature measurement capability. For these units, the STM temperature probes or sensors may be shipped already connected to the unit's I/O panel, with a pre-specified length of wire to reach to the batteries. This wire, as well as the probes or sensors at the end of the wire, is designed for the particular application, such as a formation room or a lab environment, as specified. For best results when using a probe, insert the probe at least 2 inches into the battery fluid.

The STM temperature measurement system is based on a semiconductor sensor, which may be mounted inside a probe. As a result, only original factory probes or sensors should be used since, other probes, such as thermocouples, will not work with these inputs.

3.3.4.3 Reference Voltage Inputs

Some units are equipped with optional reference voltage inputs. In this case, the voltage input leads are connected directly to the pluggable terminal strips on the voltage input PCB(s). These additional voltage sense leads can be connected to monitor individual cell voltages. If the reference voltage input is a high-impedance input, it can also be used to measure half-cell voltages. The input impedance and voltage range of these inputs is listed in the Specifications section of this manual. The reference input connections are numbered and marked for polarity. Use twisted-pair cable to reduce noise pick-up. Various length reference voltage cables are available from Bitrode Corporation.

3.3.4.4 Pressure Transducer Inputs

Some units are equipped with optional pressure transducer inputs. Due to the wide variety available, transducers are not supplied with the unit unless specified by the customer. If pressure inputs are included in this unit, the Specifications section of this manual lists the quantity, transducer type, pressure range and resolution. Typically a Bitrode supplied cable for each pressure input is included with this option. The standard pressure cable is shielded, the same length as the battery cables and not terminated at the transducer end of the cable. Electrical connections for the pressure cable to the unit are located the on the I/O panel. The pressure cable should be routed away from any sources of electromagnetic interference.

3.3.4.5 Digital or Relay I/O

Some units are equipped with optional digital or relay I/O. The I/O connections are provided at the I/O panel. If included, the standard I/O for the LCV system is digital. The standard digital inputs and outputs are designed to operate at TTL voltage levels (0 and 5VDC). These digital inputs will accept either a TTL signal or they may used as dry contact inputs, with a "closed" interpreted as a low (0V) and an "open" interpreted as a high (5V). The outputs will generate TTL voltage levels and will generate or sink currents up to 10mA per output as needed. Custom digital or relay I/O can also be specified.

3.3.4.6 Thermistor Inputs

Some units are equipped with optional thermistor inputs. In this case, the thermistor leads are connected directly to the pluggable terminal strips on the thermistor input PCB(s). These additional voltage sense leads can be connected to monitor individual cell voltages. The resistance range these inputs is listed in the Specifications section of this manual. The thermistor terminal strip(s) is labeled with input number and maximum resistance. Use twisted-pair cable to reduce noise pick-up. Various length thermistor cables are available from Bitrode Corporation.

3.4 LCV Circuit Address Selection

Each LCV circuit has a unique two digit address assigned to it. This address specifies the starting address in a block of addresses for the entire cabinet of circuits. The block size is equal to the number of circuits in the cabinet and is listed in the Specifications section of this manual. Normally this address is preset at the factory prior to shipping. However, if necessary, a cabinet's address can be changed easily. This may be necessary if new cabinets are being added to a system or if a cabinet's MPU board is being replaced.

The switches used to set the circuit address are located on the MPU PCB. The MPU PCB plug-in assembly is located in the card cage mounted inside the cabinet. The cabinet's circuit breaker should be OFF before performing any of these operations. The two rotary switches (S2 & S3) are accessed on the MPU board. Each switch selects one digit of the two digit address. The lower digit (ones) is selected with S3 and the upper digit (tens) is selected with S2. Use a small screwdriver to change the switch settings. Be sure that a cabinet is not assigned address numbers within another cabinet's address block. Carefully slide the MPU PCB plug-in assembly back in to its card cage slot.

CAUTION: The LCV's MPU PCB assembly contains static sensitive devices. Use care when touching the board to prevent static discharge.

NOTE: Assigning circuit address numbers such that multiple circuits are responding to the same address will not allow the communications network to function properly. To ensure that each circuit has a unique address, the blocks of circuit addresses assigned to different cabinets must not overlap.

3.5 Bitrode Communications Adapter (BCA) Setup

The BCA setup is defined in the BCA manual (see IM10205-01).

4.0 HARDWARE OPERATION

4.1 Initial Power Up

4.1.1 LCV Cabinet

When a cabinet is being powered up for the first time, there should not be any batteries connected. Turn on the cabinet's circuit breaker. Once powered up, the cabinet is ready for network operation following the procedures discussed in the VisuaLCN Software Operation Manual.

4.1.2 Bitrode Communications Adapter (BCA)

Power up of the BCA is defined in the BCA manual (see IM10205-01).

4.2 LCV Cabinet

4.2.1 Cabinet Description

The LCV high power cabinet contains several sub-assemblies: the isolated phase controlled power supplies, the heatsink assemblies (w/ cooling fans), the control electronic power supplies, and the control electronics (the MPU Board, Analog Boards and I/O boards housed in a card cage).

The circuits can charge or discharge at constant current, constant voltage, or constant power, within the limits listed in the Specifications section of this manual. The circuit will reach and control on these values typically within 0.4 seconds from the beginning of the test. All programming and data collection is performed with an external computer communicating over the BCA network.

The typical LCV high power circuit may be programmed to charge or discharge based on constant current, constant voltage, constant power, or any combination of the three. A charge, discharge, or rest step will continue until a programmable limit is reached. Typical limit conditions are current, voltage, power, total time, step time, ampere hours, watt hours, percent ampere hours, percent watt hours, ±dT/dt, ±dV/dt, -ΔV max, cycles or loops. Some optional limit conditions are temperature, pressure, reference voltage and digital or relay I/O. Up to ten limit checks may be programmed for each step, and a typically a program may have as many as 99 steps (as an option, a higher number of steps can be provided). A cycle test can be terminated by any of the limit checks. Test data can be sampled and stored as fast as once per second on typical LCVs by simply specifying the desired sample interval for each step (as an option, programmable 0.1 second sample rates can be provided). The data accumulated by each circuit is automatically collected by the host computer running the VisuaLCN software package.

DANGER: Only qualified personnel should work on the cabinet if power is applied to the unit.

DANGER: Do not open the cabinet doors until the input VAC power is removed. The input VAC is present at the top of the cabinet's circuit breaker and high voltages and high currents are available at non-insulated connections inside the cabinet. The cabinet/circuit components (SCRs, rectifiers, transistors, power fuses, heatsinks, circuit boards, fans, etc.) should not be replaced nor should any other repair work be performed on the power supply until the input VAC power is removed.

CAUTION: Use of an LCV circuit above the rated maximum charge and discharge voltage, which is listed in the Specifications section of this manual, can result in damage to the circuit.

4.2.2 Parallel Circuit Capability

Bitrode's LCV parallel capability allows multiple combinations of circuits (in the same cabinet) to be paralleled. Paralleling circuits is accomplished by physically paralleling the battery cables into combinations of circuits and specifying the same combinations of circuits in the VisuaLCN software. Each combination is defined as one master circuit paralleled with one or more slave circuits. By default each circuit is defined as a master if it is not attached as a slave to another master circuit.

Although controlled by a single MPU PCB, each combination of paralleled circuits is treated as an independent address on the VisuaLCN network (under the master circuit's address). Each master circuit's test profiles and data are stored separately (data is not collected for the slaves) and each combination (whether a single circuit or a master circuit with its slaves) function completely independently from each other. Note that MPU firmware code will reject profile steps if the current parameter is greater than the maximum current available for the parallel combination as defined through the VisuaLCN software.

4.2.3 Assignable Data Channel Capability (Optional)

Bitrode's optional assignable data channel capability allows data input/output channels (temperature, pressure, cell voltage, relay I/O, thermistors, etc.) to be installed in the cabinet and assigned/re-assigned to one or more circuits as needed. Each circuit is only allowed a fixed number of data channels to be assigned to itself. The maximum limit is called the circuit's data "fields" (see the Specifications section of this manual for the number of data fields per circuit allowed for this unit). For increased flexibility, each data channel can be assigned to one or more data fields on one or more circuits. These assignments can be achieved through the VisuaLCN software.

4.2.4 Isolated High Current Heatsink Description

The LCV high power cabinet has removable heatsinks, one per circuit, mounted on cooling tunnels inside the cabinet. Each LCV circuit's heatsink is isolated to allow for full flexibility when charging and discharging a battery under test. Each individual circuit's heatsink is directly controlled through a dedicated analog circuit on one of the analog PCBs in the card cage.

The heatsink consists of a phase controlled power supply (with choke/capacitor filter), an SCR switching bridge, a heatsink control PCB, a regulator heatsink (with transistors & transistor control PCB), and high current fuses. The heatsink is cooled by a fan (or fans) mounted on the cooling tunnel at the front of the heatsink. Each circuit's heatsink is connected to an isolated secondary from the main transformer. This secondary AC power is then rectified and regulated to provide the necessary DC voltage for charging/discharging its attached battery.

Also, this design allows for paralleling of the circuits on a single battery as well as connecting the circuits to different points on a series string. Note that the differential between connection points on a series

string should not exceed the maximum circuit isolation voltage (listed in the specification section of this manual).

DANGER: Only qualified personnel should work on the isolated high current heatsink if power is applied to the unit. High voltages and high currents are available at non-insulated connections inside the cabinet & heatsinks. The heatsink components (SCRs, rectifiers, transistors, power fuses, heatsinks, circuit boards, etc.) should not be replaced nor should any other repair work be performed on the heatsink with the doors open until the input VAC power is removed.

4.2.5 Controls, Indicators and Jacks

The only controls on the LCV high power cabinet are the main circuit breaker, circuit breakers for each LCV circuit and the optional Emergency Stop (E-stop) switches. The main circuit breaker provides protection to the input power mains in case of a catastrophic failure. It also acts as the main power switch applying/removing input AC power to the cabinet components. The circuit breaker for each circuit provides protection to the transformer secondaries in case of a catastrophic failure. It also acts as the circuts power switch applying/removing AC power to the circuit components. When activated, the optional E-stop switch(s) trip the circuit breaker removing power from the cabinet. If the circuit breaker is turned off while a test is running, the test will resume from where it left off as soon as the power is re-applied.

The only indicator on the LCV high power cabinet is the light on the front of each LCV cabinet identifying that AC power is applied though the main circuit breaker.

There are several indicators internal to the LCV cabinet, located on the MPU plug-in assy and on the circuit heatsink PCBs. The MPU plug-in assy has LEDs to indicate communication with the BCA network. The heatsink control PCB has LEDs to indicate +24V power, Charge, Discharge, Phase Control, Heatsink Fault & Heatsink Over-Temperature.

In the case of either fault occuring, the LCV MPU will remove current drive from the faulted circuit and identify the failure for display by the VisuaLCN client application. If upon restarting the circuit, the Over-Temperature LED continues to come on, first check for a faulty cooling fan or impedance to the airflow for that circuit. If the failure continues, then check for a thermostat failure or a Heatsink Control PCB failure. If, upon restarting the circuit, the Heatsink Fault LED continues to come on, contact Bitrode for repair of the heatsink as this indicates an over current through one or more of the heatsink transistors. Note that the transistor heatsink PCB also has a Heatsink Fault LED to help, in multi-heatsink applications, identify which transistor heatsink has faulted.

There are no test jacks on the outside of the LCV cabinet.

4.3 Battery Connection

Before a test battery is connected to a circuit, the circuit should be powered up and checked that it is not running using the external computer. When making the connections to the battery be certain to observe proper polarity. Connect both the current carrying leads and the voltage sense leads. The current leads and voltage sense leads may already be joined at the battery ends of the leads.

If your unit is equipped with optional temperature probes, pressure transducers, cell voltage, relay I/O and/or thermistor inputs, these connections should be made before running the circuit. All voltage sense connections should be clean and tight.

DANGER: Do not connect or disconnect the test battery if the circuit is running. An explosion-causing arc at the battery connections could result.

5.0 CALIBRATION & PREVENTIVE MAINTENANCE

5.1 Calibration

Before shipment from Bitrode Corporation the unit was calibrated with equipment traceable to the Nation Bureau of Standards. The LCV cabinets are calibrated via calibration software available only through Bitrode Corporation. Before calibrating, the Calibration Application portion of the VisuaLCN manual must be read and understood.

NOTE: The measurement techniques used in the LCV are so precise that the results may be affected by the presence of high-level external interference in the area where the LCV system is located. Deviations of up to 1% of full-scale readings of voltage and current may be seen under high levels of interference. To assure reliable operation in the presence of interference, the unit must be installed and operated in accordance with the instructions in this manual. The unit should be located as far away from potential interference sources as possible; in particular, sources of radio-frequency (RF) energy such as cellular phones or other portable communications equipment must not be located (during use) within three meters (ten feet) of the LCV cabinet.

5.1.1 Equipment Required

To properly calibrate an LCV circuit the equipment used should be of known/traceable calibration. The following equipment is required:

-Digital voltmeter: 4 ½ digit, with appropriate voltage ranges.

-A precision shunt for the current range of the circuit. A 4 ½ digit ammeter, with an appropriate current range, may be used instead of a shunt for any or all of the current ranges in the circuit.

-Current and voltage sense leads.

5.1.2 Calibration Procedure

After proper preparation, the Calibration Application may be started, and any circuits available through the LCN Server may be calibrated at this point. For detailed instructions on using the Calibration Application, refer to the section of the VisuaLCN manual on the Calibration Application.

The Calibration Application allows the user to control the entire calibration process, including identifying circuits and data inputs to be calibrated, initiating and carrying out the calibration sequence, and maintaining and viewing the calibration history. The calibration sequence on an individual circuit includes zeroing the voltage measurement and the current measurements for all current ranges, setting the gains on these same measurements, and verifying each gain after it is set. Depending on the type of data input channel, the calibration sequence for input channels includes calibrating the zero and/or full-scale input measurements and verifying each zero and full-scale measurement after it is set. The application and the corresponding manual will guide the user carefully through each step of this sequence.

In summary, the calibration process, to be performed by a qualified technician, is as follows:

For current calibration -

-calibrated meters and a precision shunt are externally connected to a circuit's battery cables,

-the technician selects the current range and the Calibration Application enables the default current for the specified range. The technician then measures the actual current and enters this value manually through the software interface,

-the Calibration Application downloads this value to the cabinet's MPU PCB and the MPU's embedded firmware converts this value into new calibration factors for this current range and stores them in the MPU's battery backed memory.

For voltage calibration -

-a calibrated meter and a voltage source are connected to a circuit's battery cables,

-the technician then measures the actual voltage and enters this value manually through the software interface,

-the Calibration Application downloads this value to the cabinet's MPU PCB and the MPU's embedded firmware converts this value into new voltage calibration factors and stores them in the MPU's battery backed memory.

For typical data input channel calibration (some inputs are not calibrateable) -

-an input source in connected to the data input,

-the technician then measures the source and enters this value manually through the software interface,

-the Calibration Application downloads this value to the cabinet's MPU PCB and the MPU's embedded firmware converts this value into new data input calibration factors and stores them in the MPU's battery backed memory.

This state of the art calibration mechanism allows the calibration to be accomplished quickly without accessing the control PCBs (i.e. there are no potentiometers to adjust on the control PCBs).

Note that a complete LCV system may contain several different types of circuits. The various circuits may have different current and voltage ratings and the cabinets may also have calibratable data input channels (temperature, cell voltage, pressure, thermistor, etc). When calibrating a particular type of circuit or data input channel, refer to the ratings listed in the Specifications section of this manual.

5.2 Preventive Maintenance

The LCV is designed for years of trouble-free operation, with almost no maintenance required. However, the unit should be periodically inspected and preventive maintenance performed as follows:

Monthly, or more as needed:

-On units so equipped, check all fans and filters (if applicable), to make sure that they are properly operating and free to rotate at high speed, and that air is being blown through the unit by the fan. Replace any filter that is clogged with dirt, dust, or debris.

Annually, or more as needed:

-Bolted and terminal-strip connections should be inspected for tightness; in particular, high-current connections need to be checked and tightened if necessary. The terminals that connect to the battery under test should also be inspected frequently by operators, cleaned with a wire brush or other suitable means, or replaced if they are worn to the point that a tight connection at the battery post cannot be maintained.

-External leads, and visible internal leads, should be inspected for damage to the insulation, and replaced if necessary.

-Dust buildup on parts in the forced-air path may occur in certain environments; the unit should be visually checked in these cases. Dust buildup can be removed with a soft brush or by blowing compressed air across the contaminated area.

-If dust has collected inside the airflow sensor ports (some systems do not have air flow sensors), the airflow sensor should be removed by a qualified technician and cleaned by dipping the tip end only in an ultrasonic bath. Contact Bitrode for the appropriate cleanser solution specification.

6.0 SPECIFICATIONS

The data included in this specification section has been carefully checked and is believed to be accurate. However, no responsibility is assumed for inaccuracies. Should any discrepancies or questions occur, the customer may contact Bitrode for assistance.

ENVIRONMENTAL SPECIFICATIONS

Unless otherwise noted in this section, the following environmental specifications apply to this unit:

Operating temperature range:	0 to 40° Celsius
Relative humidity:	10% to 90% RH, non-condensing
Altitude:	Sea level to 2000 meters
CE Installation Category:	II
CE Pollution Degree:	II

CABINET/CIRCUIT SPECIFICATIONS

LCV5-50-12 (MA10763-06)
w/ Parallel Circuit Capability and Assignable Data Channels
(5 Type J Temperature & 12 Hi Imp.Cell Voltage Inputs)
(Type "lD")

Hardware Configuration:

The circuits are housed in a single cabinet and each circuit contains a separate isolated phase controlled power supply. The LCV parallel capability allows multiple combinations of circuits (connected to the same card cage) to be paralleled. Each individual circuit is directly controlled through a dedicated analog circuit on one of the analog PCBs in the card cage. Assignable data channels are also supported with this unit. Each data channel can be assigned to one or more circuits through the VisuaLCN software.

Control Modes:

Charge, Discharge or Rest

Control Mode Parameters:

Current, Voltage and Power

Data Sample Parameters:

Time, Amps, Volts or Temperature

Limit Conditions:

Current, Voltage, Power, Amp-Hours, Watt-Hours, % Amp Hours Recharge, % Watt Hours Recharge, Temperature, Total Test Time, Step Time, -dV max, ±dV/dt, ±dT/dt, dV, dI, dT, Cycles, Loop1, Loop2 and Loop3

Input Power	See Serial Number Nameplate
Cabinet Dimensions (W x D x H)	56.5" x 30.6" x 77.6"
Cabinet Weight	~1500 lbs.
Circuits per Cabinet	5 Circuits
Maximum Circuits per Cabinet	5 Circuits
Circuit Address Block Size	5 Circuits per Cabinet
Data Acquisition Rate (maximum)	0.1 Second/Data Sample

LCV5-50-12 (MA10763-06) CABINET/CIRCUIT SPECIFICATIONS (continued)

Charge Voltage Range	0 to 18 Volts
Discharge Voltage Range	14 to 0 Volts
Resolution	0.001 Volts
Max. Charge/Discharge Current (per cabinet)	250 Amps
Max. Charge/Discharge Current (per circuit)	50 Amps
Resolution	0.01 Amps
Max. Charge Power (per cabinet)	3000 Watts
Max. Discharge Power (per cabinet)	3000 Watts
Max. Charge Power (per circuit)	600 Watts
Max. Discharge Power (per circuit)	600 Watts
Resolution	0.1 Watt
Fans:	5 x 120CFM (2 per Circuit)
Fuses: (115VAC: Power Modules 1-3)	1 x MDL6
(115VAC: Power Modules 4-5)	1 x MDL6
(115VAC: Card Cage Module)	1 x MDL6
(Power Module)	5 x ANN50 (2 per Circuit)
Maximum Circuit Isolation Voltage	200 VDC
Max. Battery Cable Voltage Drop	1.00 Volts

(includes total voltage drop across all negative & positive current carrying leads whether circuit is connected to cell switching cables or not.)

Assignable Data Channel Inputs:

Temperature:	Qty per Card Cage	5
	Input Type	Type J Thermocouples
	Range	-40°C to +200°C
	Resolution	0.5°C
Cell Voltage:	Qty per Card Cage	12
	Range	-8 to 8 Volts
	Resolution	0.001 Volts
	Cell Voltage Harness (4 Cell, 20ft.)	PA10832-01

LCV5-50-12 (MA10763-06) CABINET/CIRCUIT SPECIFICATIONS (continued)

DATA MAPPING DETAILS

Assignable Data Fields 20 per Circuit

Assignable Data Channels:

There are multiple input channels installed in this cabinet. Each input data channel can be assigned to a data field on a circuit within the same Card Cage. These assignments can be accomplished through the VisuaLCN software.

The 12 channels of Cell Voltage are located in I/O Slot #1.

The 5 channels of Temperature are located in I/O Slot #2.